# 11 Chemistry Unit 2 Topic Test 3: Gases I (Greenhouse Gases and pV = nRT) — Worked Solutions and Marking Guide

**TEACHER MARKING GUIDE — not for distribution to students**

Total 40 marks: Section A 10 × 1 mark (separate multiple-choice answer sheet), Section B 30 marks. Writing time 46 minutes (40 marks × 1.15 min per mark). Students use the VCE Chemistry Data Book (SLC 25 °C and 100 kPa; $V_m$ = 24.8 L mol$^{-1}$ at SLC; R = 8.31 J mol$^{-1}$ K$^{-1}$; 1 atm = 101.3 kPa). **One mark per idea or step; no half marks.** Units and significant figures are marked once, at the final answer of each part. **Rounding tolerance:** accept a final answer that differs only in the last significant figure because intermediate values were rounded; wider bands are stated where needed. **Consequential marking:** where a part uses an earlier answer, award full marks for correct working from the student's own earlier value (shown as *Consequential*). Each item is labelled with its difficulty (Easy / Medium / Hard, by marks: 10 / 20 / 10) and demand level (L1 recall, L2 routine procedure, L3 interpret or explain, L4 evaluate or justify).

## Section A — Multiple-choice answer key

Key balance: A × 2, B × 3, C × 2, D × 3 (key string D B C A D B A C B D). Every distractor is the answer produced by one named slip, so tallies of wrong letters show which misconceptions to reteach.

| Q | Key | Working for the key | Distractor diagnosis |
|---|---|---|---|
| 1 | **D** | **Easy (L2).** SLC: 100 kPa (and 25 °C). 100 ÷ 101.3 = 0.987 atm | A: thinks SLC pressure is 1 atm; B: inverted the conversion (101.3 ÷ 100); C: multiplied by 101.3 instead of dividing. |
| 2 | **B** | **Easy (L2).** T(K) = −18 + 273 = 255 K | A: ignored the minus sign; C: subtracted 273 instead of adding; D: dropped the sign and did not add 273. |
| 3 | **C** | **Easy (L1).** Greenhouse gases such as $H_2O(g)$, $CO_2$ and $CH_4$ absorb infrared radiation. | A: confuses the greenhouse effect with ozone and ultraviolet absorption; B: thinks greenhouse gases absorb incoming sunlight; D: does not know which region of the spectrum is involved. |
| 4 | **A** | **Medium (L3).** Gas particles move rapidly and randomly in all directions, so collisions (and hence force per unit area) are the same on every part of the wall. | B: thinks gas particles are stationary; C: thinks gravity sets the pressure in a small volume of gas (and contradicts 'equally'); D: thinks pressure comes from forces between particles rather than collisions with the walls. |
| 5 | **D** | **Medium (L2).** n = pV ÷ RT = 450 × 0.400 ÷ (8.31 × 295) = 0.0734 mol | A: °C not converted to K; B: volume left in mL; C: volume in m$^3$ used with pressure in kPa. |
| 6 | **B** | **Easy (L1).** Burning coal, oil and gas since the industrial revolution (land clearing also contributes) has released the extra $CO_2$. | A: names a natural source that has not increased; C: photosynthesis removes $CO_2$; D: confuses water vapour with carbon dioxide. |
| 7 | **A** | **Medium (L3).** A rise this fast and this far outside the 800 000-year natural range coincides with industrialisation; the other options do not point to a human cause. | B: true, but says nothing about the cause of the rise; C: irrelevant to the cause of the rise; D: supports a natural explanation, not a human one. |
| 8 | **C** | **Medium (L2).** p = 1.75 kPa; V = nRT ÷ p = 3.20 × 10$^{-4}$ × 8.31 × 293 ÷ 1.75 = 0.445 L = 445 mL | A: used the pressure in Pa as if it were kPa; B: °C not converted to K; D: converted L to mL twice. |
| 9 | **B** | **Hard (L3).** n($H_2$) = 2.0 ÷ 2.0 = 1.0 mol; n(He) = 2.0 ÷ 4.0 = 0.50 mol. Same V and T, so p ∝ n: p(X) = 2 × p(Y). | A: thinks pressure depends on mass, not amount (mol); C: thinks heavier particles give greater pressure (ratio inverted); D: used M = 1.0 g mol$^{-1}$ for $H_2$ (atomic H). |
| 10 | **D** | **Hard (L2).** p = 1.20 × 101.3 = 121.6 kPa; n = 121.6 × 0.750 ÷ (8.31 × 313) = 0.0351 mol; m = 0.0351 × 16.0 = 0.561 g | A: pressure left in atm; B: °C not converted to K; C: volume left in mL. |

## Section B — Short-answer worked solutions

Each part shows the worked solution, then the mark allocation (one mark per step or idea), consequential notes and answers to reject.

## Question 1 — Methane from a coal mine (9 marks)

**a)** (1 mark)

Wetlands (swamps, bogs, marshes); also accept termites or natural geological seeps.

**1 mark** a natural source: wetlands/swamps, termites, natural geological seeps, wild animals' digestion, thawing permafrost. *[Easy; L1]*

*Reject:* coal mining, livestock, rice paddies, landfill, gas leaks (all human activities).

**b)** (1 mark)

Raising livestock such as cattle and sheep; also accept rice growing, landfill (rotting waste), leaks from natural-gas wells and pipelines.

**1 mark** a human source of methane other than mining. *[Easy; L1]*

*Reject:* burning fossil fuels (releases $CO_2$, not $CH_4$); wetlands or termites (natural sources); 'pollution' alone.

**c)** (3 marks)

$T = 38.0 + 273 = 311$ K

$n = pV \div RT = (97.5 \times 1.20 \times 10^6) \div (8.31 \times 311) = 4.53 \times 10^4$ mol

$m = 4.53 \times 10^4 \times 16.0 = 7.24 \times 10^5$ g = 724 kg

**1 mark** T = 311 K used. **1 mark** $n = 4.53 \times 10^4$ mol. **1 mark** m = 724 kg (accept 723–725 kg). *[Medium; L2]*

*Reject:* $5.93 \times 10^3$ kg (°C not converted); 774 kg (used $V_m = 24.8$ L $mol^{-1}$: the gas is not at SLC); answers in g lose the last mark.

**d)** (2 marks)

Methane captured per day = $0.80 \times 724$ kg = 579 kg

Per year = $0.80 \times 724.3 \times 365 = 2.12 \times 10^5$ kg = 212 t (211 t if the daily value is rounded to 579 kg first)

**1 mark** applies both the 80% and the 365 days ($0.80 \times 724 \times 365$). **1 mark** 212 t (accept 211–212 t). *[Hard; L2]*

*Consequential:* On part c): (part c) × 0.80 × 365 ÷ 1000.

*Reject:* 52.9 t (used the 20% that is not captured); 264 t (did not apply the 80%); answers in kg when tonnes were asked lose the second mark.

**e)** (2 marks)

Methane from natural sources is part of the natural cycle that existed before human activity; it maintains the natural greenhouse effect.

The mine's methane would have stayed trapped in the coal; it reaches the atmosphere only because people mine the coal, so it raises the atmospheric $CH_4$ level above the natural level. That human-caused increase is what the enhanced greenhouse effect refers to.

**1 mark** natural-source methane belongs to the natural greenhouse effect (it was there before human activity). **1 mark** the mine's methane is released by a human activity (mining) and adds $CH_4$ above the natural level, so it counts towards the enhanced effect. *[Medium; L4]*

*Reject:* 'Methane from coal is not natural' (the stem says both formed naturally); 'mine methane is more concentrated / more harmful'; 'methane damages the ozone layer'.

*Part c) is a deliberate non-SLC trap: $V_m = 24.8$ L $mol^{-1}$ must not be used. Part e) tests natural vs enhanced: the source is natural, the release is human-caused.*

## Question 2 — Every breath you take (9 marks)

**a)** (1 mark)

37.0 + 273 = 310 K

**1 mark** 310 K. *[Easy; L2]*

*Reject:* 236 K (subtracted 273); 310 °C.

**b)** (1 mark)

101 × 1000 = 1.01 × $10^5$ Pa

**1 mark** 1.01 × $10^5$ Pa (101 000 Pa). *[Easy; L2]*

*Reject:* 0.101 Pa; 1.01 × $10^3$ Pa.

**c)** (2 marks)

V = 500 mL = 0.500 L

n = pV ÷ RT = (101 × 0.500) ÷ (8.31 × 310) = 0.0196 mol

**1 mark** correct substitution with V = 0.500 L (and kPa) or 5.00 × $10^{-4}$ $m^3$ (and Pa). **1 mark** 0.0196 mol. *[Medium; L2]*

*Consequential:* On part a) for T.

*Reject:* 19.6 mol (volume in mL); 0.0202 mol (used $V_m$ = 24.8: not at SLC).

**d)** (3 marks)

n($CO_2$) per breath = 0.040 × 0.0196 = 7.84 × $10^{-4}$ mol

Breaths per day = 12 × 60 × 24 = 17 280; n($CO_2$) per day = 17 280 × 7.84 × $10^{-4}$ = 13.5 mol

m = 13.5 × 44.0 = 596 g (0.596 kg)

**1 mark** n($CO_2$) per breath (applies the 4.0%). **1 mark** amount per day, 13.5 mol (uses 17 280 breaths). **1 mark** 596 g (accept 593–598 g, or 6.0 × $10^2$ g to 2 significant figures to match the 4.0% data). *[Hard; L3]*

*Consequential:* On part c).

*Reject:* 1.49 × $10^4$ g (4.0% not applied); 24.8 g (one hour only).

**e)** (2 marks)

The same number of particles now occupies a larger volume, so they collide with the lung walls less frequently: the pressure inside the lungs falls.

The pressure inside becomes lower than the air pressure outside, so air is pushed into the lungs until the pressures are equal.

**1 mark** larger volume → less frequent collisions with the walls → lower pressure inside. **1 mark** inside pressure lower than outside (atmospheric), so air moves in from higher to lower pressure. *[Medium; L3]*

*Reject:* ‘The lungs suck air in’ with no pressure difference; ‘the particles slow down’ (temperature is unchanged).

*Parts a) and b) are routine conversions; part d) chains percentage, rate and mass.*

## Question 3 — Heating a classroom (6 marks)

**a)** (1 mark)

The pascal, Pa.

**1 mark** pascal (Pa). *[Easy; L1]*

*Reject:* kPa (accept only if ‘pascal’ is also given); atm; ‘newtons’.

**b)** (2 marks)

Heating makes the air particles move faster, so they collide with the walls more frequently and with more force.

For the pressure to stay at 101 kPa in the same volume, fewer particles can be present, so some air escapes through gaps around doors and windows.

**1 mark** particles move faster: more frequent and harder collisions with the walls. **1 mark** pressure stays at 101 kPa with the volume fixed, so the number of particles must decrease: air leaves the room. *[Medium; L3]*

*Reject:* 'The air particles expand'; 'hot air rises' with no particle reasoning.

**c)** (3 marks)

n at 278 K = $(101 \times 2.40 \times 10^5) \div (8.31 \times 278) = 1.049 \times 10^4$ mol

n at 295 K = $(101 \times 2.40 \times 10^5) \div (8.31 \times 295) = 9.89 \times 10^3$ mol

$\Delta n = 605$ mol; $m = 605 \times 28.8 = 1.74 \times 10^4$ g = 17.4 kg

**1 mark** n at 5.0 °C (278 K). **1 mark** n at 22.0 °C (295 K). **1 mark** mass leaving = 17.4 kg (accept 17–18 kg, allowing for rounding of the two amounts). *[Hard; L3]*

*Reject:* °C used (division by 5.0 and 22.0 gives a very different answer); 302 kg (mass of all the air at 5.0 °C, not the mass that leaves); answers in g lose the last mark.

*Part c) subtracts two similar numbers, so rounding the amounts early shifts the answer: hence the wide tolerance.*

## Question 4 — Commuting by car (6 marks)

**a)** (1 mark)

$40 \times 168 = 6720$ g = 6.72 kg

**1 mark** 6.72 kg. *[Easy; L2]*

*Reject:* 6720 kg; 4.2 kg.

**b)** (3 marks)

$n(CO_2) = 6720 \div 44.0 = 153$ mol

T = 16.0 + 273 = 289 K

$V = nRT \div p = 153 \times 8.31 \times 289 \div 100.5 = 3.65 \times 10^3$ L

**1 mark** n = 153 mol. **1 mark** T = 289 K used. **1 mark** $V = 3.65 \times 10^3$ L (accept $3.64–3.66 \times 10^3$ L). *[Medium; L2]*

*Consequential:* On part a).

*Reject:* $3.79 \times 10^3$ L (used $V_m = 24.8$: not at SLC); 202 L (°C not converted).

**c)** (2 marks)

Larger.

At constant pressure and amount, V is proportional to the kelvin temperature ($V = nRT \div p$). 200 °C = 473 K, which is higher than 289 K, so the volume is larger (about 1.6 times). In particle terms, faster particles hit the surroundings harder, so the gas expands until its pressure matches the outside.

**1 mark** larger (committed judgement). **1 mark** reason: $V \propto T$ (in K) at constant p and n, or particle explanation (faster particles push the gas out to a larger volume at the same pressure). *[Medium; L3]*

*Reject:* 'Smaller, because gases contract when hot'; 'the same, because the mass is the same'.

*Part c) connects the gas equation to the particle model without a calculation.*

## Mark summary

| Section | Marks | Notes |
|---|---|---|
| A — multiple choice | 10 | 10 × 1 mark; key A2 / B3 / C2 / D3 |
| B — short answer | 30 | Q1 9, Q2 9, Q3 6, Q4 6 |
| **Total** | **40** | Writing time 46 minutes |

## Difficulty tally (by marks)

| Section | Easy | Medium | Hard | Total |
|---|---|---|---|---|
| A — multiple choice | 4 | 4 | 2 | 10 |
| B — short answer | 6 | 16 | 8 | 30 |
| **Total** | **10** | **20** | **10** | **40** |
| Target (25 / 50 / 25%) | 10 | 20 | 10 | 40 |

## Difficulty and demand by item

| Item | Marks | Difficulty | Demand |
|---|---|---|---|
| A Q1 | 1 | Easy | L2 |
| A Q2 | 1 | Easy | L2 |
| A Q3 | 1 | Easy | L1 |
| A Q4 | 1 | Medium | L3 |
| A Q5 | 1 | Medium | L2 |
| A Q6 | 1 | Easy | L1 |
| A Q7 | 1 | Medium | L3 |
| A Q8 | 1 | Medium | L2 |
| A Q9 | 1 | Hard | L3 |
| A Q10 | 1 | Hard | L2 |
| B Q1a) | 1 | Easy | L1 |
| B Q1b) | 1 | Easy | L1 |
| B Q1c) | 3 | Medium | L2 |
| B Q1d) | 2 | Hard | L2 |
| B Q1e) | 2 | Medium | L4 |
| B Q2a) | 1 | Easy | L2 |
| B Q2b) | 1 | Easy | L2 |
| B Q2c) | 2 | Medium | L2 |
| B Q2d) | 3 | Hard | L3 |
| B Q2e) | 2 | Medium | L3 |
| B Q3a) | 1 | Easy | L1 |
| B Q3b) | 2 | Medium | L3 |
| B Q3c) | 3 | Hard | L3 |
| B Q4a) | 1 | Easy | L2 |
| B Q4b) | 3 | Medium | L2 |
| B Q4c) | 2 | Medium | L3 |

## Cognitive demand

| Demand level | Marks | Share |
|---|---|---|
| L1 recall | 5 | 12% |
| L2 routine procedure | 18 | 45% |
| L3 interpret / explain | 15 | 38% |
| L4 evaluate / justify | 2 | 5% |

Routine marks (L1 + L2): 23 of 40 (58%).

## Key knowledge assessed

• $CO_2$, $CH_4$ and $H_2O$ as greenhouse gases: infrared absorption, natural and enhanced greenhouse effect, sources

• gas pressure as particle collisions; SLC

• pV = nRT with unit conversions (kPa, Pa, atm, mL, L, °C, K)

Not assessed (not taught this year): indicators and indicator choice, pH calculations, back titrations, calibration curves, solubility curves, colorimetry, gravimetric analysis.